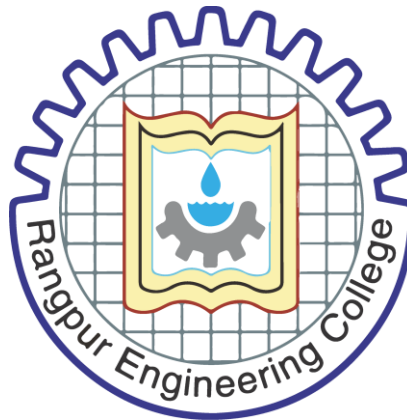


Rangpur Engineering College, Rangpur
Department of Computer Science and Engineering
(Affiliated by University of Rajshahi)



WeavePDF

A Project Proposal
Course code: CSE 4192
Course Title: Thesis/ Project (Part I)

Submitted By

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WeavePDF

All-in-One PDF & Productivity Web Platform

Project Proposal

Project Title	WeavePDF: All-in-One PDF & Productivity Web Platform
Project Type	Full-Stack Web Application (With NextJS & MongoDB)
Target Users	Students, Professionals, Businesses, Remote Teams
Proposed Stack	Next.js, React, Node.js, MongoDB, JWT, Tailwind CSS

1. Introduction

In today's digital-first world, professionals, students, and businesses frequently work with documents across multiple platforms. WeavePDF is proposed as a comprehensive browser-based platform that unifies PDF management, document conversion, collaboration, and productivity tools into a single secure environment. The platform aims to simplify document workflows, improve efficiency, and provide additional premium features through a subscription-based model.

- **All-in-One Document Management:** PDF editing, document conversion, file organization, and essential productivity tools in one platform.
- **Collaboration & Smart Utilities:** Real-time collaboration, speech-to-text conversion, quiz generator, resume builder, and integrated communication tools.
- **Premium Services & Secure Payments:** Dedicated premium features with a secure digital payment gateway for seamless access to advanced tools and services.

2. Problem Statement

At present, users who need to work with PDF documents and related productivity tools are forced to navigate across many separate platforms. A student may use one website to convert a Word document to PDF, another to add a watermark, a third to generate a quiz from study material, and yet another tool for collaboration. Each switch causes friction, wastes time, and introduces security risks when sensitive documents are uploaded to unfamiliar third-party services.

The key problems this project addresses are:

- Lack of a unified platform combining PDF tools, collaboration, and productivity features.
- Privacy concerns when uploading sensitive documents to multiple third-party services.

- No seamless payment integration for accessing premium document management services.
- Difficulty in building CVs, resumes, or quizzes without switching between multiple tools.
- Limited accessibility users without strong technical backgrounds find it hard to manage complex PDF workflows.
- No built-in communication or collaboration layer within document management platforms.

3. Objectives

The primary objective of WeavePDF is to create a comprehensive and user-friendly document management platform that simplifies PDF-related tasks and enhances productivity. The platform aims to combine essential document tools, collaboration features, and premium services into a single secure and scalable web application.

The primary objectives of WeavePDF are:

- To develop a fully functional, responsive web application for PDF management and document productivity.
- To integrate important features including speech-to-text, quiz generation from PDFs, and PDF-to-audiobook conversion.
- To implement a secure digital payment gateway for premium feature access and service monetization.
- To provide real-time collaboration through an integrated messenger chat system.
- To offer document personalization tools including signature drawing, watermarking, and page numbering.
- To build resume and CV builders that help users create professional documents without external tools.
- To deliver an admin dashboard with insights, user management, and download tracking.
- To ensure a secure, scalable, and user-friendly experience using modern full-stack technologies.
- Unlock advanced tools and exclusive functionality through premium subscription plans.

4. Key Features

WeavePDF offers a wide range of integrated tools designed to simplify document management, enhance productivity, and improve collaboration. These features combine traditional PDF utilities with modern AI-powered solutions, creating a complete all-in-one platform for personal, academic, and professional use.

4.1 Digital Payment Gateway

A secure and smooth payment processing system will be integrated to support premium tool access. Users will be able to subscribe to plans or make one-time payments using trusted

payment providers, with transaction history and billing management available from the dashboard.

4.2 Merge & Split PDF

Users can combine multiple PDF files into a single document or divide a large PDF into smaller, more manageable parts. This module ensures document organization remains easy and efficient for both personal and professional use.

4.3 Edit PDF

A rich PDF editor will allow users to add text, highlight sections, insert images, apply watermarks, rotate pages, and draw digital signatures directly within the browser without the need for any desktop software.

4.4 Word to PDF & Markdown to PDF

Seamless document conversion from Microsoft Word (.docx) and Markdown (.md) formats to PDF, preserving all formatting, styles, and structure during conversion.

4.5 Speech to Text

An AI-powered speech recognition module will allow users to convert spoken words into editable text in real time. This feature is particularly useful for note-taking, transcription, and hands-free content creation.

4.6 Image to PDF

Users can upload one or multiple images (JPG, PNG, etc.) and convert them into a well-formatted, shareable PDF document. This is useful for digitizing notes, receipts, and photos.

4.7 Draw Signature & Add Watermark

Users can draw digital signatures on a canvas and embed them into PDF documents. A watermarking tool allows branding or protection of documents with text or image overlays.

4.8 Add Page Numbers

A dedicated module for adding page numbers to PDF documents will ensure that long reports, assignments, and official papers remain well-structured and easy to navigate.

4.9 Todo List

A built-in task management tool will allow users to create, organize, and track their tasks directly within the platform — keeping work and document management tightly integrated.

4.10 Messenger/WhatsApp Chat

A real-time messaging interface will support team collaboration, allowing users to discuss documents, share files, and coordinate tasks without leaving the platform.

4.11 PDF to Audiobook

An accessibility-focused feature that converts PDF documents into spoken audio, making content more consumable for users on the move or those with reading difficulties.

4.12 Admin Dashboard

A centralized dashboard for administrators to monitor platform usage, manage user accounts, track recent downloads, and view performance insights through visual analytics.

4.13 Resume & CV Builder

Interactive, template-based builders that allow users to create professional resumes and CVs by filling in structured forms, with the option to export them as polished PDF documents.

4.14 Quiz Generator

An AI-powered module that analyzes uploaded PDFs and automatically generates multiple-choice or short-answer quizzes ideal for students, educators, and corporate trainers.

4.15 QR Code Generator

A simple and fast QR code generation tool that converts URLs, text, or document links into downloadable QR codes, useful for sharing and marketing purposes.

5. Technologies to be Used

The development of WeavePDF will leverage modern web technologies to ensure high performance, scalability, security, and maintainability. These technologies have been selected to provide a seamless user experience while supporting advanced document processing and collaboration features.

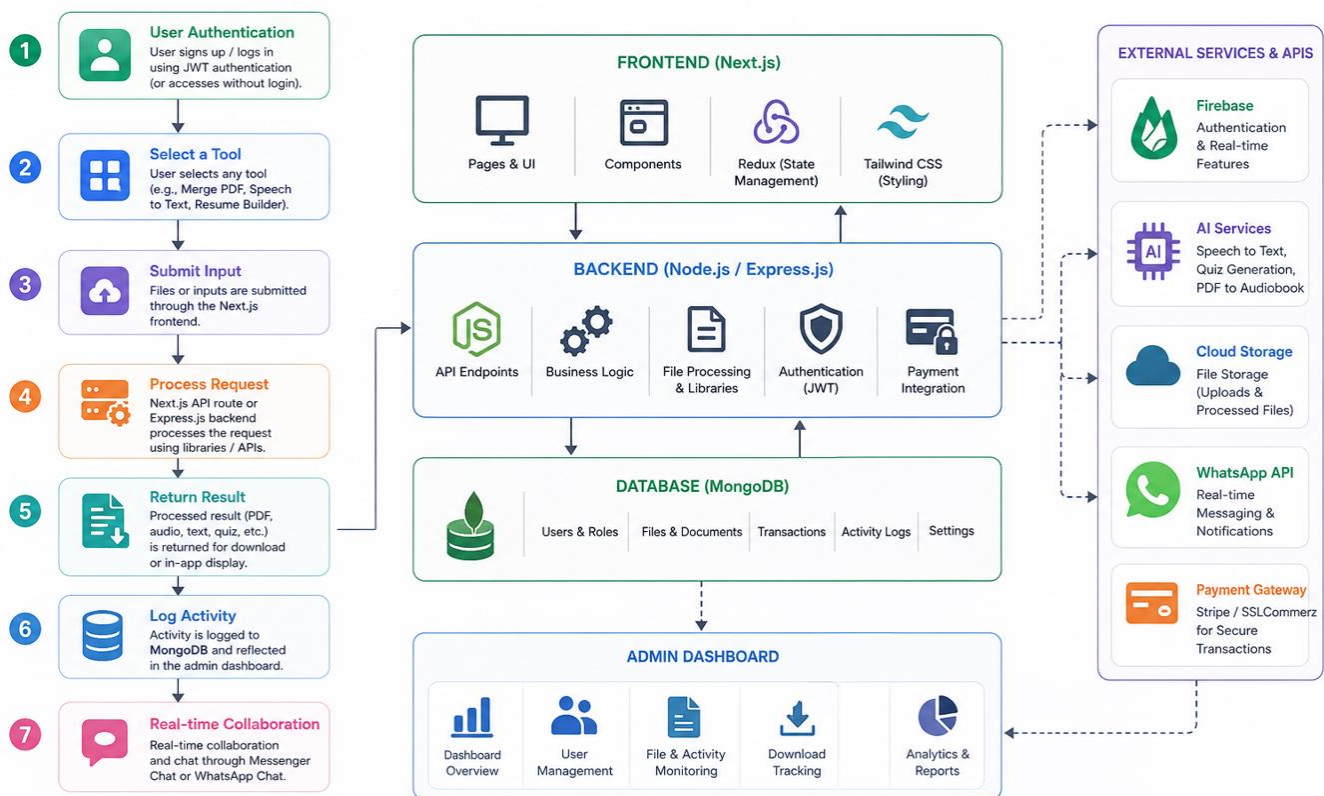
Component	Technology
Frontend Framework	Next.js (App Router)
Language	JavaScript, TypeScript
State Management	Redux Toolkit
Styling	Tailwind CSS, HTML5, CSS3
Backend	Node.js, Express.js
Authentication	Json Web Token
Database	MongoDB with Mongoose
Payment Integration	Stripe or SSLCommerz
PDF Processing	Npm packages
Version Control	Git & GitHub
API Testing	Postman
Deployment	Vercel (Frontend), Vercel (Backend)

6. System Architecture & Workflow

WeavePDF follows a modern client-server architecture built on Next.js for both the frontend and server-side rendering layer, with a dedicated Node.js/Express.js API for backend processing. The platform uses MongoDB as the primary data store, Firebase for authentication and real-time features, and integrates external APIs for AI capabilities.

The general workflow for document processing follows these steps:

- User authenticates via JWT and lands on the home page. OR user can access without login.
- User selects a tool (e.g., Merge PDF, Speech to Text, Resume Builder).
- Files or inputs are submitted through the NextJS frontend.
- The Next.js API route or Express.js backend processes the request, calling relevant libraries or external APIs.
- The processed result (PDF, audio, text, quiz) is returned to the user for download or in-app display.
- Activity is logged to MongoDB and reflected in the admin dashboard.
- Real-time collaboration and chat will be done with messenger chat or WhatsApp chat.



7. Modules Breakdown

The WeavePDF platform is organized into multiple functional modules, each responsible for handling specific features and services. This modular architecture improves maintainability, scalability, and overall system performance while ensuring a seamless user experience.

Module	Description
Authentication Module	Handles user registration, login, JWT sessions, and role-based access using user credentials.
Payment Module	Integrates Stripe or SSLCommerz for secure subscription and one-time payment processing.
PDF Processing Module	Core engine for merge, split, edit, watermark, signature, and page number operations using npm packages.
Conversion Module	Converts Word and Markdown documents to PDF using Puppeteer and markdown-pdf.
Node packages	Drives speech-to-text, quiz generation from PDFs, and PDF-to-audiobook features via npm packages.
Image & QR Module	Handles image-to-PDF conversion and dynamic QR code generation.
Document Builder Module	Powers the interactive Resume and CV builder with downloadable PDF output.
Collaboration Module	Provides real-time messenger chat and WhatsApp directly.
Todo Module	Task management system with CRUD operations stored in MongoDB.
Dashboard & Admin Module	Displays usage analytics, manages users, and tracks file downloads for administrators.

8. Expected Outcomes

- A fully functional, deployed web application accessible from any modern browser.
- A complete PDF management suite covering merge, split, edit, convert, watermark, and signature tools.
- Important tools for speech-to-text, quiz generation, and PDF-to-audiobook conversion.
- Secure digital payment processing with subscription and billing management.
- An intuitive admin dashboard with real-time insights and user management.
- A built-in messenger or WhatsApp for real-time team collaboration within the platform.
- Professional resume and CV builders with PDF export capability.
- Improved productivity for students, professionals, and organizations by eliminating tool-switching.

9. Project Timeline (Tentative)

The development of WeavePDF will follow a structured timeline to ensure efficient implementation, testing, and deployment of all features. Each phase focuses on a specific aspect of the platform, enabling systematic progress and timely project completion.

Phase / Task	Duration
Requirement Analysis & Planning	1 week
UI/UX Design (Figma Prototypes)	2 weeks
Project Setup & Authentication	1 week
PDF Core Module Development	2 weeks
Conversion & AI Features	2 weeks
Payment Gateway Integration	1 week
Resume/CV Builder & Quiz Generator	2 weeks
Messenger/WhatsApp, Todo & Dashboard	2 weeks
Admin Panel Development	1 week
Testing & Bug Fixing	2 weeks
Deployment & Documentation	1 week
Total Estimated Duration	~17 weeks (Approximately)

10. Conclusion

WeavePDF is proposed as a modern, full-featured, and user-centered SaaS platform designed to consolidate the fragmented world of digital document management and productivity tools. By combining PDF processing, AI capabilities, real-time collaboration, secure payments, and personal productivity features into a single, beautifully designed application, WeavePDF aims to save users time, enhance security, and deliver measurable improvements in daily workflow efficiency.

Built on a robust and industry-standard technology stack including Next.js, React, Node.js, MongoDB, JWT, and Tailwind CSS the platform is designed to be scalable, maintainable, and extensible. The first release will focus on delivering the core PDF and productivity features, with a clear roadmap for AI enhancements, mobile support, and enterprise-level capabilities in future versions. WeavePDF has strong academic value as a comprehensive full-stack engineering project and significant real-world commercial potential as an independent SaaS product serving students, businesses, and professionals globally.